Adaptive Quant Portfolio System

Machine Learning-Based Regime Detection and ETF Allocation

Steve Fernandes

April 2026 – May 2026

Overview

This project develops an adaptive quantitative portfolio system that uses machine learning-based market regime detection to dynamically allocate across exchange-traded funds (ETFs). The objective is not simply to maximize absolute return, but to improve risk-adjusted performance by reducing volatility and drawdowns during unfavorable market environments.

The system uses historical ETF price data, engineers financial features such as momentum, volatility, trend, and drawdown indicators, applies K-Means clustering to identify market regimes, and assigns portfolio weights based on the detected regime. The final strategy is backtested against a passive SPY buy-and-hold benchmark.

Although the adaptive strategy produced a lower CAGR than SPY, it achieved a higher Sharpe ratio, lower annual volatility, and smaller maximum drawdown. This suggests that the strategy was more effective from a risk-management perspective.

1 Project Objective

Traditional buy-and-hold investing maintains fixed exposure regardless of market conditions. However, financial markets shift between different regimes, including bullish periods, defensive periods, and high-volatility environments.

The objective of this project is to build a systematic portfolio framework that:

- Detects market regimes using unsupervised machine learning.
- Allocates dynamically across ETFs based on market conditions.
- Uses financial features related to momentum, volatility, trend, and drawdown.
- Backtests the adaptive strategy against SPY.
- Evaluates performance using CAGR, Sharpe ratio, maximum drawdown, and volatility.

2 ETF Universe and Data

The strategy uses five ETFs representing different market exposures.

ETF	Role in Portfolio
SPY	U.S. equity market exposure
QQQ	Technology and growth equity exposure
TLT	Long-term U.S. Treasury bond exposure
GLD	Gold / defensive alternative asset exposure
SHY	Short-term Treasury bond exposure

Table 1: ETF universe used in the adaptive allocation system.

Historical ETF price data was downloaded using Python and the `yfinance` package beginning in 2015. The prices were transformed into daily returns and used to construct the backtest.

3 Feature Engineering

The regime detection model uses features derived from SPY market behavior. These features are intended to summarize momentum, volatility, trend, and downside risk.

3.1 Momentum Features

The strategy calculates short-term and medium-term returns:

$$r_{21d} = \frac{P_t}{P_{t-21}} - 1$$

$$r_{63d} = \frac{P_t}{P_{t-63}} - 1$$

These features capture recent market strength or weakness over approximately one-month and three-month windows.

3.2 Volatility Features

Rolling volatility is calculated over 21-day and 63-day windows:

$$\sigma_{21d} = \text{Std}(r_{t-21}, ..., r_t)$$

$$\sigma_{63d} = \text{Std}(r_{t-63}, ..., r_t)$$

Higher volatility may indicate unstable or defensive market environments.

3.3 Trend Feature

The model uses a 50-day and 200-day moving average trend signal:

$$\text{Trend}_{50,200} = \frac{MA_{50}}{MA_{200}} - 1$$

A positive value suggests that the shorter-term trend is stronger than the long-term trend.

3.4 Drawdown Feature

Drawdown measures the decline from the previous peak:

$$\text{Drawdown}_t = \frac{P_t}{\max(P_1, ..., P_t)} - 1$$

This feature captures downside market stress.

4 Machine Learning Methodology

The project uses K-Means clustering for unsupervised market regime detection. Before clustering, the features are standardized using `StandardScaler` so that each variable contributes fairly to the clustering process.

The K-Means model is specified as:

$$K = 3$$

The three resulting clusters are interpreted as:

- Bullish regime
- Defensive regime
- High-volatility regime

The model does not use pre-labeled regimes. Instead, it groups market environments based on similarities in return, volatility, trend, and drawdown behavior.

5 Regime Interpretation

After clustering, each regime is interpreted based on the average values of the engineered features.

Cluster	21d Ret.	63d Ret.	21d Vol.	63d Vol.	Trend	Drawdown
0	0.0179	0.0592	0.1140	0.1198	0.0595	-0.0174
1	-0.0601	-0.0686	0.2714	0.2258	-0.0038	-0.1411
2	0.0559	0.0141	0.2066	0.2720	-0.0296	-0.0882

Table 2: Average feature values by market regime cluster.

Cluster 0 is interpreted as a bullish regime due to positive returns, lower volatility, positive trend, and limited drawdown. Cluster 1 is interpreted as defensive due to negative returns, high drawdown, and elevated volatility. Cluster 2 is interpreted as high volatility because it shows elevated volatility and weaker trend conditions.

6 Portfolio Allocation Rules

Portfolio weights are assigned based on the detected regime. The allocation rules are designed to increase equity exposure during bullish markets and shift toward defensive assets during riskier environments.

- In bullish regimes, the strategy allocates more heavily toward SPY and QQQ.
- In defensive regimes, the strategy increases exposure to TLT, GLD, and SHY.
- In high-volatility regimes, the strategy balances risk assets with defensive ETFs.

The portfolio return is calculated as:

$$R_{p,t} = \sum_{i=1}^{n} w_{i,t-1} R_{i,t}$$

where $w_{i,t-1}$ is the lagged portfolio weight for ETF i, and $R_{i,t}$ is the ETF return at time t. Lagged weights are used to reduce look-ahead bias.

7 Backtesting Framework

The strategy is backtested from 2015 onward. The process is:

1. Download ETF price data.
2. Calculate ETF returns.
3. Engineer market features from SPY.
4. Standardize features.
5. Apply K-Means clustering to identify market regimes.
6. Map clusters to regime labels.
7. Assign ETF weights based on each regime.
8. Calculate strategy returns using lagged weights.
9. Compare performance against SPY buy-and-hold.

8 Performance Metrics

The strategy is evaluated using four key metrics.

8.1 Compound Annual Growth Rate

$$CAGR = (1 + R_{\text{total}})^{\frac{1}{T}} - 1$$

8.2 Sharpe Ratio

$$\text{Sharpe Ratio} = \frac{\sqrt{252} \cdot \bar{r}}{\sigma_r}$$

8.3 Maximum Drawdown

$$\text{Max Drawdown} = \min \left(\frac{V_t}{\max(V_1, ..., V_t)} - 1 \right)$$

8.4 Annualized Volatility

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} \sqrt{252}$$

9 Results

The adaptive strategy was compared against a passive SPY buy-and-hold benchmark.

Metric	**Adaptive Strategy**	**SPY Benchmark**
CAGR	11.57%	14.98%
Sharpe Ratio	1.09	0.91
Maximum Drawdown	-26.67%	-32.05%
Annual Volatility	10.57%	17.09%

Table 3: Performance comparison between the adaptive strategy and SPY.

The SPY benchmark generated a higher compound annual growth rate. However, the adaptive strategy produced a stronger Sharpe ratio, lower maximum drawdown, and significantly lower volatility.

10 Interpretation of Results

The results suggest that the adaptive strategy is more effective as a risk-managed allocation framework than as a pure return-maximization strategy.

SPY achieved a higher CAGR of 14.98%, compared to 11.57% for the adaptive strategy. However, SPY also experienced higher annualized volatility of 17.09%, compared to 10.57% for the adaptive system. This means the adaptive strategy generated returns with less variability.

The adaptive strategy also had a smaller maximum drawdown of -26.67%, compared to -32.05% for SPY. This indicates that the adaptive allocation helped reduce downside exposure during periods of market stress.

Most importantly, the adaptive strategy achieved a Sharpe ratio of 1.09, compared to 0.91 for SPY. This shows that the adaptive system delivered better risk-adjusted performance even though it produced lower absolute return.

11 Key Findings

- The adaptive strategy did not outperform SPY on absolute return.
- The strategy outperformed SPY on risk-adjusted return as measured by the Sharpe ratio.
- The strategy reduced annualized volatility from 17.09% to 10.57%.
- The strategy reduced maximum drawdown from -32.05% to -26.67%.
- K-Means clustering provided a simple but effective framework for separating market environments.
- Dynamic allocation was more useful for risk control than for maximizing total return.

12 Limitations

This project is a research prototype and has several limitations:

- Transaction costs and slippage are not fully modeled.
- K-Means clustering assumes fixed cluster structure and may not adapt perfectly to future regimes.
- The portfolio allocation rules are manually defined.
- The backtest may be sensitive to ETF selection and feature lookback windows.
- The strategy uses historical relationships that may not persist in future markets.

13 Future Improvements

Future extensions could include:

- Adding transaction costs and slippage.
- Using walk-forward validation.
- Comparing K-Means with Hidden Markov Models or Gaussian Mixture Models.
- Testing different ETF universes.
- Optimizing weights using minimum variance or maximum Sharpe optimization.
- Adding macroeconomic features such as interest rates, inflation, and credit spreads.
- Performing out-of-sample testing over multiple market cycles.

14 Conclusion

This project demonstrates how machine learning and quantitative finance techniques can be combined to build a dynamic ETF allocation system. By engineering financial indicators, detecting market regimes with K-Means clustering, and applying regime-based portfolio weights, the strategy provides a systematic approach to risk-managed investing.

While the adaptive strategy did not beat SPY in absolute return, it delivered better risk-adjusted performance by achieving a higher Sharpe ratio, lower volatility, and smaller drawdowns. This makes the project especially relevant to quantitative risk management, portfolio construction, and systematic investing.